

SAT-Style Practice Test - DSAT Mar 2025 INT v3 W

Original practice version for Reading and Writing plus Math. The passages, contexts, names, data, values, and answer choices are newly created.

SECTION 1, MODULE 1: Reading and Writing

Question 1

Skill: Words in Context

During her four-decade career, materials scientist Lina Ortega published studies on ceramics, biodegradable foams, and solar-cell coatings. Reviewers often note the remarkable breadth of her work: her research was not confined to a single specialty but was instead _____ across several branches of applied science.

Which choice completes the text with the most logical and precise word or phrase?

- A) vacant
- B) varied
- C) obsolete
- D) isolated

Answer: B

Question 2

Skill: Textual Evidence - Inference

The following text is adapted from a novel. The narrator is preparing for a school recitation.

I had promised myself I would deliver the speech exactly as printed, every comma in its place. Yet when I stood beneath the auditorium lights, the middle paragraph vanished from my mind. I kept talking anyway, inventing a bridge from one remembered sentence to the next. The teachers smiled, but I knew I had been improvising, not reciting.

Based on the text, when the narrator says she had been "improvising," what does she most likely mean?

- A) She was speaking too softly for the audience to hear.
- B) She was translating the speech into another language.
- C) She was refusing to continue after forgetting a paragraph.
- D) She was creating parts of the speech as she performed it.

Answer: D

Question 3

Skill: Words in Context

The Urban Bird Archive is a crowdsourced project that asks residents to upload photographs and field notes about birds seen in their neighborhoods. Volunteers review these submissions so that unusual sightings, migration dates, and nesting locations can be added as verified _____ to the archive.

Which choice completes the text with the most logical and precise word or phrase?

- A) entries
- B) excuses
- C) rivalries
- D) predictions

Answer: A

Question 4

Skill: Words in Context

Innovation districts are often assumed to form because start-up companies, universities, and investors _____ one another's goals. Sociologist Priya Menon argues, however, that some districts emerge mainly because the groups share roads, cafes, and suppliers, not because they necessarily agree on long-term priorities.

Which choice completes the text with the most logical and precise word or phrase?

- A) withdraw from
- B) stumble upon

- C) align with
- D) take over

Answer: C

Question 5

Skill: Central Ideas and Details

The following text is adapted from a short story.

When I arrived at Aunt Mira's house, she had already moved a desk into the spare room and placed a stack of blank notebooks beside the window. I had meant to stay only two nights, long enough to attend the festival and leave. But Aunt Mira listened to my vague plan, asked no questions that would embarrass me, and then said that the notebooks would be there whether I used them or not. By evening, my temporary visit had begun to feel like permission to remain.

Which choice best states the main purpose of the text?

- A) It explains why the narrator dislikes attending local festivals.
- B) It shows how Aunt Mira's quiet preparation helps the narrator feel welcome.
- C) It describes the process by which Aunt Mira renovates a room.
- D) It argues that written invitations are more effective than spoken ones.

Answer: B

Question 6

Skill: Text Structure and Purpose

Community archaeology invites local residents to help professional archaeologists document historic sites. This approach can preserve knowledge that might otherwise be missed, such as oral histories about abandoned buildings or traditional uses of artifacts. It was central to a recent survey of coastal fishing villages, in which researchers combined satellite images with interviews from residents whose families had lived in the area for generations.

Which choice best describes the overall structure of the text?

- A) It presents two competing theories and evaluates the evidence for each theory.
- B) It describes a problem, rejects one solution, and proposes an unrelated solution.
- C) It introduces a historical figure and then summarizes that figure's career.
- D) It defines a research approach, explains benefits of the approach, and gives an example of its use.

Answer: D

Question 7

Skill: Central Ideas and Details

The island of Paloma is one of the smaller moons orbiting the planet Giras. It was first detected in 1988 by astronomers using a ground-based telescope. Later, images from the Arcadia space probe revealed two additional moons in 2004, and a 2019 survey identified five more. As of 2024, scientists had confirmed forty-three moons around Giras.

According to the text, in which year was Paloma first detected?

- A) 1988
- B) 2004
- C) 2019
- D) 2024

Answer: A

Question 8

Skill: Comparing Texts

Founded in 1982, the Harbor Textile Museum displays woven goods from communities around the Indian Ocean and has more than 6,000 items in its collection. Since 2010, several museums devoted to textile art have opened elsewhere. One notable example is the Loom House in Cape Town, which opened in 2018 and focuses on contemporary South African

weaving.

Which contrast does the text draw between the Harbor Textile Museum and the Loom House?

- A) The Loom House has a larger collection.
- B) The Harbor Textile Museum focuses only on contemporary art.
- C) The Loom House opened more recently.
- D) The Harbor Textile Museum is located in Cape Town.

Answer: C

Question 9

Skill: Command of Evidence - Literary

In the 1905 novel *The House of Mirth*, Edith Wharton often presents rooms and objects as signals of social status. A student claims that Wharton sometimes depicts luxury as beautiful but also faintly oppressive.

Which quotation from *The House of Mirth* would most effectively illustrate the claim?

- A) "The train arrived at noon, and several passengers stepped quickly onto the platform."
- B) "The drawing-room glittered with lamps, yet the heavy curtains seemed to shut out every breath of air."
- C) "She folded the letter and placed it inside the drawer beside her gloves."
- D) "The garden path curved toward a small gate at the far end of the lawn."

Answer: B

Question 10

Skill: Command of Evidence - Supporting Claims

A team of engineers developed a biodegradable film that can be wrapped around produce to slow moisture loss. The film is made from seaweed extract and citrus fiber. Engineer Tomas Ibarra claims that although the film performs well in laboratory tests, producing it at commercial scale may be difficult.

Which finding, if true, would most directly support Ibarra's claim?

- A) The film can be adjusted to wrap fruits of many different shapes.
- B) Seaweed extract and citrus fiber are both safe for contact with food.
- C) The film breaks down in soil faster than several petroleum-based plastics do.
- D) The seaweed extract needed for the film is available only in small seasonal batches.

Answer: D

Question 11

Skill: Command of Evidence - Textual Support

In a memoir, travel writer Aria Sen directly addresses the reader while explaining a difficult custom from her childhood village.

Which quotation from Sen's memoir would most effectively illustrate the claim?

- A) "You may wonder, dear reader, why no one entered the courtyard before sunrise."
- B) "My uncle kept the keys on a brass ring that hung near the kitchen door."
- C) "We walked along the river until the market stalls came into view."
- D) "The first rains arrived two weeks later than the farmers had hoped."

Answer: A

Question 12

Skill: Data Analysis - Weakening a Claim

[Graph: A bar graph shows average annual export growth rates for three countries before and after joining regional trade agreements. Before: Norland 4%, Costa Verde 9%, Meridia -2%. After: Norland 2%, Costa Verde 13%, Meridia 12%.]

A student claims that joining a regional trade agreement always increased a country's export growth rate in the five years

after joining.

Which choice best describes data from the graph that weaken the student's claim?

- A) Costa Verde's export growth rate was higher after joining than before joining.
- B) Meridia had negative export growth before joining but positive growth after joining.
- C) Norland's export growth rate was lower after joining than before joining.
- D) All three countries had positive export growth after joining.

Answer: C

Question 13

Skill: Data Analysis - Supporting a Conclusion

[Table: Average time lost per employee from work interruptions after two workplace programs began. Focus-training group: 6 weeks = 154 minutes, 12 months = 102 minutes. Email-policy group: 6 weeks = 160 minutes, 12 months = 221 minutes.]

Researchers tested whether focus training reduced workplace interruptions more effectively than a new email policy. They concluded that focus training became more effective over time.

Which choice best describes data from the table that support the researchers' conclusion?

- A) At 6 weeks, the focus-training group lost more time than the email-policy group.
- B) At 6 weeks, the two groups had similar average time lost, but by 12 months the focus-training group lost much less time than the email-policy group.
- C) Both groups lost exactly the same amount of time at 12 months.
- D) The email-policy group lost less time at 12 months than it had lost at 6 weeks.

Answer: B

Question 14

Skill: Command of Evidence - Supporting an Underlined Claim

Born in Senegal in 1923, musician Sira Diop helped popularize mbalax fusion, a genre that combined traditional sabar drumming with jazz and electric guitar. Diop toured widely across West Africa and Europe in the 1960s. Her work created a foundation for later musicians who adapted local rhythms to address modern urban life. As the movement spread beyond Senegal, the range of musical traditions incorporated into mbalax fusion also expanded.

Which detail about later mbalax fusion songs, if true, would best illustrate the underlined claim?

- A) Many were recorded in studios that used newer microphones than those available in the 1960s.
- B) Many were shorter than the songs Diop performed during live concerts.
- C) Many were released on vinyl before being reissued digitally.
- D) Many included rhythms from Ghanaian highlife and Malian blues as well as Senegalese drumming.

Answer: D

Question 15

Skill: Standard English Conventions - Verbs

The City Atlas Project, a public history website, invites residents _____ photographs and short descriptions of buildings that no longer exist.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) to submit
- B) submits
- C) submitting
- D) submitted

Answer: A

Question 16

Skill: Standard English Conventions - Verbs

Next month, the museum _____ a restored 1920s film projector that technicians have spent three years repairing.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) displays
- B) displayed
- C) will display
- D) has displayed

Answer: C

Question 17

Skill: Standard English Conventions - Boundaries

Capoeira, a Brazilian art form that combines dance, music, and martial movement, has become an international _____ the first global youth capoeira festival was held in Lisbon in 2015.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) phenomenon and
- B) phenomenon, and
- C) phenomenon. And
- D) phenomenon, in

Answer: B

Question 18

Skill: Standard English Conventions - Boundaries

Red Mesa Park, which covers nearly 90,000 acres, is part of the Aravaipa _____ located in southern Arizona.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Preserve, it is
- B) Preserve. Is
- C) Preserve, is
- D) Preserve and is

Answer: D

Question 19

Skill: Standard English Conventions - Punctuation

The documentary River Lines examines how maps of the Mississippi changed during the nineteenth _____ showing how floods, trade, and politics influenced cartographers' choices.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) century,
- B) century;
- C) century
- D) century and,

Answer: A

Question 20

Skill: Standard English Conventions - Punctuation

For centuries, many historians assumed that the fortress had been abandoned suddenly. This was not the _____ recent excavations show that the site was left gradually over several decades.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) case however;

- B) case; however
- C) case, however,
- D) case, however

Answer: C

Question 21

Skill: Transitions

A survey asked participants to name animals they associated with patience. Tortoises were ranked first, and elephants were ranked second. _____, cats were ranked ninth, behind owls and whales.

Which choice completes the text with the most logical transition?

- A) Specifically,
- B) On the other hand,
- C) Therefore,
- D) Later,

Answer: B

Question 22

Skill: Transitions

In 1927, surveyors estimated the geographic center of the province using a weighted paper map and a balancing pin. _____, when a computer model recalculated the center almost a century later, the new estimate differed by less than two kilometers.

Which choice completes the text with the most logical transition?

- A) Fittingly,
- B) Similarly,
- C) In other words,
- D) Still,

Answer: D

Question 23

Skill: Transitions

A ring of debris inside a planet's Roche limit is expected to break apart rather than gather into a moon. Astronomers recently observed a compact object orbiting well outside the calculated Roche limit of its dwarf planet; _____, the object has remained intact for many years.

Which choice completes the text with the most logical transition?

- A) fittingly,
- B) for example,
- C) likewise,
- D) though,

Answer: A

Question 24

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Slow cinema features long, uninterrupted shots of ordinary actions.
- The 2020 film *Window Repair* shows a craftsperson restoring a wooden frame in real time.
- Some critics call slow cinema tedious.
- Film scholar Nora Hale writes that the genre can offer viewers "a precise attention to everyday labor."
- The student wants to use a quotation to challenge the claim that slow cinema is tedious.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Slow cinema can be tedious because it often presents ordinary actions in real time.
- B) Window Repair is a 2020 film that shows a craftsman restoring a wooden frame.
- C) While some critics call slow cinema tedious, Hale argues that it can offer viewers "a precise attention to everyday labor."
- D) Nora Hale is a film scholar who has written about slow cinema.

Answer: C

Question 25

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Fossil specimens are housed in natural history museums around the world.
- Many specimens are given nicknames.
- The North Valley Museum houses a stegosaur specimen nicknamed Lantern.
- Lantern lived during the Late Jurassic period.
- The student wants to provide an example of a fossil specimen's nickname.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Many natural history museums house fossils from the Late Jurassic period.
- B) Lantern is the nickname of a stegosaur fossil specimen housed at the North Valley Museum.
- C) Fossil specimens can be found in museums around the world.
- D) The North Valley Museum contains many fossils, including some with nicknames.

Answer: B

Question 26

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- A soundscape is the collection of sounds heard in a particular place.
- In 2009, Chile's Ministry of Culture created a list of important soundscapes.
- Each soundscape was chosen for cultural significance.
- The calls of sea lions at Valdivia Harbor are on the list.
- Valdivia Harbor is in southern Chile.
- The student wants to specify where a culturally significant soundscape can be heard.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) A soundscape can include animal calls heard in a particular place.
- B) Chile's Ministry of Culture selected each listed soundscape for cultural significance.
- C) The list created in 2009 included many culturally significant soundscapes.
- D) The calls of sea lions at Valdivia Harbor can be heard in southern Chile.

Answer: D

Question 27

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Many plant species on Madeira are either native or introduced.
- In a 2021 study, researchers investigated how birds disperse seeds on the island.
- The laurel tree *Laurus novocanariensis* is native to Madeira.
- The shrub *Pittosporum undulatum* is introduced to Madeira.
- Seeds from both plants were found in bird droppings.
- The student wants to emphasize a difference between the two plants.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Although *Laurus novocanariensis* and *Pittosporum undulatum* both grow on Madeira, only the former is native there.

- B) Birds on Madeira can disperse seeds from both native and introduced plants.
- C) Researchers in 2021 studied bird droppings to learn about seed dispersal on Madeira.
- D) *Pittosporum undulatum* seeds and laurel seeds were both found in bird droppings.

Answer: A

SECTION 1, MODULE 2: Reading and Writing

Question 1

Skill: Words in Context

Prizes for scientific achievement are sometimes awarded to a single researcher. In other cases, a committee recognizes work completed by two or three people together; then the award is given _____, as when a 2022 mathematics prize honored two collaborators for a shared theorem.

Which choice completes the text with the most logical and precise word or phrase?

- A) accidentally
- B) reluctantly
- C) jointly
- D) secretly

Answer: C

Question 2

Skill: Words in Context

Like many river deltas, the Arvand Delta is _____: its channels shift, islands form and disappear, and the shoreline changes shape as tides and sediment flows interact.

Which choice completes the text with the most logical and precise word or phrase?

- A) dynamic
- B) sustainable
- C) unraveled
- D) immutable

Answer: A

Question 3

Skill: Words in Context

Many early travel posters showed beaches and mountains in exaggeratedly romantic colors. By contrast, photographer Etta Rowe's later images of seaside towns seemed _____; their straight horizons and unposed figures gave the impression of neutral documentation.

Which choice completes the text with the most logical and precise word or phrase?

- A) tranquil
- B) peculiar
- C) nostalgic
- D) dispassionate

Answer: D

Question 4

Skill: Text Structure and Purpose

The following text is adapted from a nineteenth-century short story.

By the standards of his town, Mr. Arden was admired as a man of polished manners and uncommon learning. His rare visits to the capital, however, earned him a different reputation. There, where his elaborate phrases sounded less like wisdom than performance, people began to suspect that he spoke not to illuminate a subject but to conceal how little he understood it. Mr. Arden sensed the suspicion and responded with even grander speeches, which only confirmed it.

Which choice best describes the overall structure of the text?

- A) It describes Mr. Arden's childhood, lists his major accomplishments, and compares him with another scholar.

- B) It presents a favorable local view of Mr. Arden, contrasts that view with a less favorable view elsewhere, and explains how his behavior reinforces the less favorable view.
- C) It argues that small towns are always more honest than capitals and then provides evidence for that claim.
- D) It explains why Mr. Arden decides to leave the capital permanently and return home.

Answer: B

Question 5

Skill: Text Structure and Purpose

Architect Amina Desai's career, which stretched from the 1930s to the 1980s, developed in distinct phases. After studying in Cairo and Paris, Desai began incorporating principles of modernism into her work. The River School, designed after her return to Mumbai, shows this shift clearly: its unadorned concrete forms differ sharply from the decorative arches seen in her earliest houses.

Which choice best describes the overall structure of the text?

- A) It disputes a common interpretation of an architect's work and replaces it with a new theory.
- B) It describes one building's materials and then explains how those materials were manufactured.
- C) It summarizes an architect's career, identifies a transition in that career, and illustrates the transition with a specific building.
- D) It compares two architects and argues that one influenced the other.

Answer: C

Question 6

Skill: Text Structure and Purpose

Researchers Sofia Beltran, Hui Chen, and Omar Haddad created a computer model to predict which combinations of clinics, grocery stores, transit stops, and parks best serve a neighborhood. Because neighborhood boundaries can be subjective, the model defines a neighborhood as a cluster of places that residents are likely to visit during a typical week. The researchers suggest that city planners could use the model to identify areas lacking essential services.

Which choice best describes the overall structure of the text?

- A) It introduces a research team's model, explains one part of its method, and suggests a possible use for the model.
- B) It lists several unrelated services and argues that parks are the most important.
- C) It describes a failed experiment and then proposes that the experiment be abandoned.
- D) It compares rural and urban neighborhoods and concludes that they have identical needs.

Answer: A

Question 7

Skill: Cross-Text Connections

Text 1

Some astronomers define the habitable zone as the range of distances from a star where a rocky planet could maintain liquid water on its surface. A 2021 model by Leila Vargas placed the inner edge of the habitable zone for certain red-dwarf systems at 0.18 astronomical units (AU). Vargas therefore described the planet KX-47b, located at 0.19 AU, as a strong candidate for retaining surface water.

Text 2

In 2024, Martin Osei and Priya Rao argued that planets around some red dwarfs can experience intense stellar flares that strip away atmospheres. Their model placed the effective inner edge of the habitable zone farther out, at about 0.27 AU for systems like KX-47. They therefore cautioned against assuming that KX-47b can retain liquid water.

Based on the texts, how would Osei and Rao (Text 2) most likely respond to Vargas's assessment in Text 1?

- A) By agreeing that KX-47b is a strong candidate because it lies beyond 0.18 AU
- B) By claiming that red-dwarf systems cannot contain any rocky planets
- C) By stating that liquid water is unnecessary when identifying habitable-zone planets
- D) By arguing that Vargas's model may place the inner edge of the habitable zone too close to the star

Answer: D

Question 8

Skill: Central Ideas and Details

When novelist Talia Mensah, who had previously published in English, began writing fiction in Twi in the 1980s, some critics warned that the decision would narrow her audience. Mensah disagreed, arguing that using her first language would let her represent local humor, idioms, and family relationships more precisely. Her later books won major awards and have been translated into more than twenty languages.

Which choice best states the main idea of the text?

- A) Mensah stopped writing fiction after her early books were translated into English.
- B) Mensah's decision to write in Twi was questioned by some critics but ultimately contributed to her literary success.
- C) Critics generally believed that Twi was more widely understood than English.
- D) Mensah's later books avoided local humor and family relationships.

Answer: B

Question 9

Skill: Central Ideas and Details

In a study of residents of Quito, Ecuador, and Rosario, Argentina, researchers asked whether participants used city parks. Of 710 respondents from Quito, 85% reported using city parks; of 690 respondents from Rosario, 70% did. However, 88% of Quito respondents and 91% of Rosario respondents reported having access to at least one park within walking distance. Thus, the higher rate of park use in Quito cannot be explained by Quito residents having greater access to nearby parks.

Which choice best describes the main idea of the text?

- A) Park use was identical in Quito and Rosario because access to nearby parks was identical.
- B) Rosario residents had much less access to nearby parks than Quito residents did.
- C) Although a greater share of Quito residents reported using parks, that difference is not due to greater access to nearby parks in Quito.
- D) The researchers concluded that increasing park access in Quito would reduce park use.

Answer: C

Question 10

Skill: Data Analysis - Interpreting Graphs

[Graph: A line graph tracks three types of kelp forest coverage from 2015 to 2022. Giant kelp rises from 12 to 18 hectares from 2015 to 2018, then falls to 9 hectares by 2022. Bull kelp rises from 8 to 16 hectares from 2015 to 2019, then falls to 7 hectares by 2022. Total kelp coverage rises from 28 to 42 hectares from 2015 to 2019, then falls to 19 hectares by 2022.]

Researchers note that kelp coverage increased for several years but then declined sharply after 2019.

Which statement, if true, would account for the data shown and illustrate the researchers' point?

- A) A marine heat wave beginning in 2020 damaged multiple kelp species across the bay.
- B) Giant kelp was never present in the bay before 2015.
- C) Bull kelp grows only in areas where giant kelp is absent.
- D) Total kelp coverage remained constant from 2015 to 2022.

Answer: A

Question 11

Skill: Command of Evidence - Supporting Claims

Piezoelectric floor tiles convert pressure from footsteps into electrical energy. Although a single step produces little power, engineer Anika Bose claims that a busy train station could use such tiles to supply meaningful energy for low-power devices such as information displays.

Which finding, if true, would most directly support Bose's claim?

- A) The tiles are more expensive to install than ordinary floor tiles.
- B) Most passengers do not notice whether a floor contains piezoelectric tiles.
- C) Piezoelectric materials are also used in some microphones.
- D) A pilot installation in a station generated enough daily electricity to operate several digital signs.

Answer: D

Question 12

Skill: Logical Completion

Like many solitary bee species, blue orchard bees face threats from habitat loss, pesticides, and climate variation. However, blue orchard bees have broader diets than some specialist bees and can nest in a variety of small cavities. Ecologist Mara Jain argues that these generalist traits are linked to resilience under changing environmental conditions. Jain therefore suggests that _____

Which choice most logically completes the text?

- A) blue orchard bees are the only insects affected by pesticide use.
- B) blue orchard bees may be less vulnerable to some environmental changes than bee species with narrower diets and nesting needs.
- C) specialist bees are likely to increase in number whenever habitat loss occurs.
- D) studying blue orchard bees cannot provide any information about bee populations.

Answer: B

Question 13

Skill: Logical Completion

Microbial fuel cells (MFCs) generate electricity when bacteria transfer electrons to an electrode. In one experiment, researchers added copper nanoparticles to the electrode surface. Because copper conducts electricity well, the researchers hypothesized that the nanoparticles increased power output by making it easier for electrons to reach the electrode. This hypothesis would be strengthened if _____

Which choice most logically completes the text?

- A) the bacteria stopped metabolizing nutrients after the nanoparticles were added.
- B) the electrode produced the same power output with and without nanoparticles.
- C) the copper nanoparticles allowed electrons to move more directly from the bacterial biofilm to the electrode.
- D) copper nanoparticles prevented the formation of any bacterial biofilm.

Answer: C

Question 14

Skill: Logical Completion

The methane mole fraction in Titan's lower atmosphere influences cloud formation and rainfall patterns. A climate model of Titan matches some observations of rainfall but assumes a single methane mole fraction throughout the lowest atmospheric layer. Planetary scientist Hana Seok notes that actual measurements are sparse and vary by altitude. It is therefore important to remember that _____

Which choice most logically completes the text?

- A) the model's agreement with some rainfall observations does not prove that its assumed methane distribution is accurate at every altitude.
- B) methane cannot influence weather on Titan because Titan is not Earth.
- C) no model can ever be useful unless it includes every possible atmospheric gas.
- D) rainfall observations are irrelevant to understanding Titan's atmosphere.

Answer: A

Question 15

Skill: Standard English Conventions - Punctuation

Included in the exhibition Material Echoes was a sculpture by artist Lorna _____ work combines found metal, fabric, and projected light.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Sayeed whose
- B) Sayeed. Whose
- C) Sayeed, which
- D) Sayeed, whose

Answer: D

Question 16

Skill: Standard English Conventions - Punctuation

Editors of the new field guide asked novelist Ian Nakatani to write the entry for "hollow," a term describing a low area in a forest. For "ridge," however, the editors chose _____ author of Mountain Weather.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) geologist Mara Singh the
- B) geologist Mara Singh, the
- C) geologist, Mara Singh, the
- D) geologist Mara Singh; the

Answer: B

Question 17

Skill: Standard English Conventions - Boundaries

Researcher Celia Dominguez studies how coral larvae respond to sound. Her latest paper examines whether reef noise can guide larvae toward suitable _____ it also considers how boat traffic may interfere with that process.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) habitats and
- B) habitats,
- C) habitats, and
- D) habitats; and

Answer: C

Question 18

Skill: Standard English Conventions - Possessives

The theater company invited three playwrights to revise _____ scripts during a weeklong workshop.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) their
- B) they
- C) them
- D) theirs

Answer: A

Question 19

Skill: Standard English Conventions - Verbs

The new battery design, along with two earlier prototypes, _____ in the engineering lab's safety report.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are discussed
- B) have discussed

- C) discussing
- D) is discussed

Answer: D

Question 20

Skill: Standard English Conventions - Boundaries

The composer wrote the piece in 1911. _____ it was not performed publicly until nearly twenty years later.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) However
- B) However,
- C) However;
- D) However and

Answer: B

Question 21

Skill: Transitions

The earliest known recipe for almond soup appears in a fifteenth-century cookbook. _____, a similar recipe appears in a sixteenth-century manuscript from a neighboring region, suggesting that the dish circulated widely.

Which choice completes the text with the most logical transition?

- A) Nevertheless,
- B) Instead,
- C) Similarly,
- D) For this reason,

Answer: C

Question 22

Skill: Transitions

The satellite was designed to operate for three years. _____, it continued sending usable data for more than a decade.

Which choice completes the text with the most logical transition?

- A) In fact,
- B) By contrast,
- C) For example,
- D) Meanwhile,

Answer: A

Question 23

Skill: Transitions

Some desert plants avoid heat by opening their stomata only at night. _____, many water lilies open their stomata during the day because their floating leaves remain relatively cool.

Which choice completes the text with the most logical transition?

- A) Therefore,
- B) Likewise,
- C) Specifically,
- D) Conversely,

Answer: D

Question 24

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The wooden church in Kerimaki, Finland, can seat thousands of people.
- It was completed in 1847.
- Local builders used pine from nearby forests.
- The student wants to emphasize the church's large capacity.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Completed in 1847, the wooden church in Kerimaki was built with pine from nearby forests.
- B) The wooden church in Kerimaki, Finland, is notable for being able to seat thousands of people.
- C) Local builders used pine to construct a church in Finland in the nineteenth century.
- D) The church in Kerimaki is made of wood and was completed in 1847.

Answer: B

Question 25

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- Botanist Mei Arakawa studied alpine mosses.
- In 2016, she published a field guide to mosses of the Japanese Alps.
- The guide includes photographs, maps, and notes on habitat.
- The student wants to introduce Arakawa and her area of study.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Japanese Alps contain mosses that appear in photographs and maps.
- B) A field guide published in 2016 includes notes on habitat.
- C) Botanist Mei Arakawa studied alpine mosses and published a field guide on them in 2016.
- D) Mosses can be studied through photographs, maps, and habitat notes.

Answer: C

Question 26

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- A palimpsest is a manuscript page from which writing has been scraped away so the page can be reused.
- Modern imaging technology can reveal erased writing.
- A university library recently used multispectral imaging on a medieval palimpsest.
- The imaging revealed a previously unknown medical text.
- The student wants to explain why modern imaging can be useful for studying palimpsests.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Because modern imaging can reveal erased writing, it can help scholars recover texts hidden in palimpsests.
- B) A palimpsest is a reused manuscript page, and some palimpsests are medieval.
- C) A university library used multispectral imaging on one medieval manuscript page.
- D) The medical text revealed by imaging was previously unknown.

Answer: A

Question 27

Skill: Rhetorical Synthesis

While researching a topic, a student has taken the following notes:

- The painter Alma Thomas worked as a teacher for many years.
- After retiring, she devoted more time to painting.
- Her abstract paintings often use bright blocks of color.
- In 1972, she became the first Black woman to have a solo exhibition at the Whitney Museum of American Art.
- The student wants to emphasize a major achievement in Thomas's career.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Alma Thomas taught for many years before spending more time on painting.
- B) Alma Thomas's abstract paintings often include bright blocks of color.
- C) Alma Thomas was both a teacher and a painter who made abstract art.
- D) In 1972, Alma Thomas became the first Black woman to have a solo exhibition at the Whitney Museum of American Art.

Answer: D

SECTION 2, MODULE 1: Math

Question 1

Skill: Linear Equations in One Variable

If $3x + 7 = 28$, what is the value of x ?

- A) 5
- B) 7
- C) 9
- D) 11

Answer: B

Solution: Subtract 7 from both sides: $3x = 21$. Divide by 3: $x = 7$.

Question 2

Skill: Percentages

A jacket originally cost \$80. During a sale, the price was reduced by 25%. What was the sale price?

- A) \$20
- B) \$55
- C) \$60
- D) \$75

Answer: C

Solution: 25% of 80 is 20, so the sale price is $80 - 20 = 60$ dollars.

Question 3

Skill: Ratios and Proportions

A recipe uses 5 cups of flour for every 2 cups of sugar. If 15 cups of flour are used, how many cups of sugar are needed?

- A) 6
- B) 4
- C) 5
- D) 8

Answer: A

Solution: The flour amount is multiplied by 3, so the sugar amount is also multiplied by 3: $2 \times 3 = 6$.

Question 4

Skill: Linear Functions

The function f is defined by $f(x) = 4x - 9$. What is $f(6)$?

- A) 18
- B) 24
- C) 33
- D) 15

Answer: D

Solution: $f(6) = 4(6) - 9 = 24 - 9 = 15$.

Question 5

Skill: Systems of Linear Equations

In the xy -plane, the lines $y = 2x + 1$ and $y = -x + 10$ intersect at the point (a, b) . What is the value of a ?

- A) 2
- B) 3

C) 4

D) 5

Answer: B

Solution: Set $2x + 1 = -x + 10$. Then $3x = 9$, so $x = 3$. Therefore $a = 3$.

Question 6

Skill: Interpreting Linear Models

A gym charges a one-time registration fee of \$30 plus \$18 per month. Which equation gives the total cost C , in dollars, for m months?

A) $C = 30m + 18$

B) $C = 48m$

C) $C = 18m + 30$

D) $C = 18(m + 30)$

Answer: C

Solution: The fixed fee is 30 and the monthly cost is $18m$, so $C = 18m + 30$.

Question 7

Skill: Quadratic Equations

What is one solution to $x^2 - 8x + 15 = 0$?

A) 3

B) 2

C) 8

D) 15

Answer: A

Solution: Factor $x^2 - 8x + 15$ as $(x - 3)(x - 5)$. Solutions are $x = 3$ and $x = 5$. One solution is 3.

Question 8

Skill: Area

A rectangle has length 14 inches and width 9 inches. What is its area, in square inches?

A) 23

B) 46

C) 98

D) 126

Answer: D

Solution: Area = length $\times$ width = $14 \times 9 = 126$ square inches.

Question 9

Skill: Data Analysis - Mean

The numbers 4, 7, 9, 10, and 15 have an average of 9. If 15 is added to the list, what is the new average?

A) 9

B) 10

C) 11

D) 12

Answer: B

Solution: Original sum = $5 \times 9 = 45$. New sum = $45 + 15 = 60$. New average = $60/6 = 10$.

Question 10

Skill: Exponential Growth

A population of bacteria doubles every hour. If there are 150 bacteria at 1:00 p.m., how many bacteria are there at 4:00 p.m.?

- A) 300
- B) 450
- C) 1,200
- D) 900

Answer: C

Solution: From 1 p.m. to 4 p.m. is 3 hours, or 3 doublings. $150 \times 2^3 = 150 \times 8 = 1,200$.

Question 11

Skill: Equivalent Expressions

Which expression is equivalent to $6(x - 4) + 2x$?

- A) $8x - 24$
- B) $8x - 4$
- C) $6x - 22$
- D) $12x - 24$

Answer: A

Solution: Distribute: $6(x - 4) + 2x = 6x - 24 + 2x = 8x - 24$.

Question 12

Skill: Linear Inequalities

Which of the following values of x satisfies $5x - 6 > 19$?

- A) 3
- B) 4
- C) 5
- D) 6

Answer: D

Solution: $5x - 6 > 19$ gives $5x > 25$, so $x > 5$. Among the choices, only 6 satisfies the inequality.

Question 13

Skill: Geometry - Angles

Two lines intersect. One angle measures 132 degrees. What is the measure, in degrees, of an adjacent angle?

- A) 42
- B) 48
- C) 58
- D) 132

Answer: B

Solution: Adjacent angles formed by intersecting lines are supplementary. $180 - 132 = 48$ degrees.

Question 14

Skill: Probability

A bag contains 6 red marbles, 5 blue marbles, and 4 green marbles. If one marble is selected at random, what is the probability that it is blue?

- A) $\frac{1}{5}$
- B) $\frac{1}{4}$
- C) $\frac{1}{3}$

D) $\frac{2}{5}$

Answer: C

Solution: There are $6 + 5 + 4 = 15$ marbles total. The probability of blue is $\frac{5}{15} = \frac{1}{3}$.

Question 15

Skill: Linear Equations in Two Variables

The equation $2x + 3y = 24$ is graphed in the xy -plane. What is the y -intercept of the graph?

A) 8

B) 6

C) 12

D) 24

Answer: A

Solution: Set $x = 0$: $3y = 24$, so $y = 8$. The y -intercept is 8.

Question 16

Skill: Function Notation

If $g(x) = x^2 + 2x$, what is $g(-3)$?

A) -15

B) -3

C) 15

D) 3

Answer: D

Solution: $g(-3) = (-3)^2 + 2(-3) = 9 - 6 = 3$.

Question 17

Skill: Circle Geometry

A circle has radius 6 centimeters. What is its circumference, in centimeters?

A) 6π

B) 12π

C) 18π

D) 36π

Answer: B

Solution: Circumference $= 2\pi r = 2\pi(6) = 12\pi$.

Question 18

Skill: Student-Produced Response - Linear Equations

If $7k - 5 = 37$, what is the value of k ?

Answer: 6

Solution: Add 5 to both sides: $7k = 42$. Divide by 7: $k = 6$.

Question 19

Skill: Student-Produced Response - Percentages

A number is increased by 20% to become 84. What was the original number?

Answer: 70

Solution: Let n be the original number. $1.20n = 84$, so $n = 84/1.20 = 70$.

Question 20

Skill: Student-Produced Response - Systems

If $x + y = 18$ and $x - y = 4$, what is the value of x ?

Answer: 11

Solution: Add the equations: $2x = 22$, so $x = 11$.

Question 21

Skill: Student-Produced Response - Geometry

A triangle has base 16 and height 9. What is its area?

Answer: 72

Solution: Area = $(1/2)(\text{base})(\text{height}) = (1/2)(16)(9) = 72$.

Question 22

Skill: Student-Produced Response - Exponents

If $2^a = 32$, what is the value of a ?

Answer: 5

Solution: $32 = 2^5$, so $a = 5$.

SECTION 2, MODULE 2: Math

Question 1

Skill: Linear Functions - Slope

Line l passes through the points $(2, 7)$ and $(8, 19)$. What is the slope of line l ?

- A) 1
- B) 3
- C) 2
- D) 4

Answer: C

Solution: Slope = $(19 - 7)/(8 - 2) = 12/6 = 2$.

Question 2

Skill: Quadratic Functions

The function f is defined by $f(x) = x^2 - 6x + 11$. What is the minimum value of $f(x)$?

- A) 1
- B) 2
- C) 3
- D) 11

Answer: B

Solution: The vertex occurs at $x = -b/(2a) = 6/2 = 3$. $f(3) = 9 - 18 + 11 = 2$.

Question 3

Skill: Nonlinear Equations

If $x > 0$ and $x^2 = 49$, what is the value of $x + 5$?

- A) 2
- B) 7
- C) 54
- D) 12

Answer: D

Solution: Since $x > 0$ and $x^2 = 49$, $x = 7$. Then $x + 5 = 12$.

Question 4

Skill: Systems of Equations

A cafe sold 12 small drinks and 8 large drinks for a total of \$76. A small drink costs \$4. What is the cost, in dollars, of one large drink?

- A) 3.50
- B) 3
- C) 4
- D) 5

Answer: A

Solution: The 12 small drinks cost $12 \times 4 = 48$ dollars. The 8 large drinks cost $76 - 48 = 28$ dollars. Each large drink costs $28/8 = 3.50$ dollars.

Question 5

Skill: Rational Expressions

For x not equal to 0, which expression is equivalent to $(12x^3 - 8x^2)/(4x^2)$?

- A) $3x^2 - 2x$

- B) $12x - 8$
- C) $3x - 2$
- D) $4x - 2$

Answer: C

Solution: Divide each term by $4x^2$: $12x^3/(4x^2) = 3x$ and $-8x^2/(4x^2) = -2$. The result is $3x - 2$.

Question 6

Skill: Geometry - Similar Triangles

Triangle A is similar to triangle B. A side of triangle A has length 10, and the corresponding side of triangle B has length 15. If another side of triangle A has length 14, what is the length of the corresponding side of triangle B?

- A) 18
- B) 21
- C) 19
- D) 24

Answer: B

Solution: The scale factor from A to B is $15/10 = 1.5$. The corresponding side is $14 \times 1.5 = 21$.

Question 7

Skill: Statistics - Median

The data set 12, 14, 18, 21, 25, 30, 32 has median m . If 40 is added to the data set, what is the new median?

- A) 19.5
- B) 21
- C) 25
- D) 23

Answer: D

Solution: With 8 values, the median is the average of the 4th and 5th values: $(21 + 25)/2 = 23$.

Question 8

Skill: Function Interpretation

A ball is thrown upward. Its height h , in meters, t seconds after it is thrown is given by $h(t) = -5t^2 + 20t + 2$. What is the height of the ball when $t = 3$?

- A) 17
- B) 20
- C) 22
- D) 47

Answer: A

Solution: $h(3) = -5(9) + 20(3) + 2 = -45 + 60 + 2 = 17$.

Question 9

Skill: Linear Models - Intercepts

A tank contains 240 liters of water and is drained at a constant rate of 12 liters per minute. Which equation gives the amount A of water, in liters, remaining after t minutes?

- A) $A = 240 + 12t$
- B) $A = 12 - 240t$
- C) $A = 240 - 12t$
- D) $A = 12t - 240$

Answer: C

Solution: The starting amount is 240 and it decreases by 12 liters per minute, so $A = 240 - 12t$.

Question 10

Skill: Trigonometry

In a right triangle, angle A has opposite side length 9 and hypotenuse length 15. What is $\sin(A)$?

- A) $\frac{4}{5}$
- B) $\frac{3}{5}$
- C) $\frac{5}{3}$
- D) $\frac{5}{4}$

Answer: B

Solution: $\sin(A) = \text{opposite/hypotenuse} = \frac{9}{15} = \frac{3}{5}$.

Question 11

Skill: Polynomial Factors

Which of the following is a factor of $x^2 + 2x - 24$?

- A) $x - 6$
- B) $x + 8$
- C) $x + 12$
- D) $x - 4$

Answer: D

Solution: $x^2 + 2x - 24 = (x + 6)(x - 4)$. Therefore $x - 4$ is a factor.

Question 12

Skill: Exponential Functions

The value of an investment is modeled by $V(t) = 500(1.06)^t$, where t is the number of years after the investment is made. What does 500 represent in this model?

- A) The initial value of the investment
- B) The yearly percent increase
- C) The value after 6 years
- D) The number of years required to double

Answer: A

Solution: When $t = 0$, $V(0) = 500(1.06)^0 = 500$. Thus 500 is the initial value.

Question 13

Skill: Circle Equation

In the xy -plane, a circle has equation $(x - 4)^2 + (y + 3)^2 = 25$. What is the radius of the circle?

- A) 3
- B) 4
- C) 5
- D) 25

Answer: C

Solution: The standard form is $(x - h)^2 + (y - k)^2 = r^2$. Here $r^2 = 25$, so $r = 5$.

Question 14

Skill: Algebra - Absolute Value

What are the solutions to $|x - 2| = 9$?

- A) -9 and 9
- B) -7 and 11

C) -11 and 7

D) 2 and 9

Answer: B

Solution: $x - 2 = 9$ or $x - 2 = -9$. Thus $x = 11$ or $x = -7$.

Question 15

Skill: Advanced Systems

If $2x + y = 13$ and $y = x^2 - 2$, which of the following could be the value of x ?

A) -4

B) -3

C) 5

D) 3

Answer: D

Solution: Substitute $y = x^2 - 2$ into $2x + y = 13$: $2x + x^2 - 2 = 13$, so $x^2 + 2x - 15 = 0$. Factor: $(x + 5)(x - 3) = 0$. The choices include 3, so the matching choice is 3.

Question 16

Skill: Volume

A cylinder has radius 3 inches and height 10 inches. What is the volume of the cylinder, in cubic inches?

A) 90π

B) 30π

C) 60π

D) 300π

Answer: A

Solution: Volume = $\pi r^2 h = \pi(3^2)(10) = 90\pi$.

Question 17

Skill: Data Analysis - Scatterplots

[Scatterplot: Points show that as the number of hours a machine operates increases from 1 to 8, its maintenance cost generally increases from about \$12 to about \$70.] Which statement best describes the association shown?

A) Negative association

B) No association

C) Positive association

D) Constant association

Answer: C

Solution: As operating hours increase, maintenance cost generally increases, so the association is positive.

Question 18

Skill: Student-Produced Response - Quadratic Equations

If $x^2 - 10x + 24 = 0$ and x is greater than 5, what is the value of x ?

Answer: 6

Solution: Factor: $x^2 - 10x + 24 = (x - 4)(x - 6)$. Solutions are 4 and 6. Since $x > 5$, $x = 6$.

Question 19

Skill: Student-Produced Response - Linear Functions

A line has slope 5 and passes through the point (2, 17). What is the y-intercept of the line?

Answer: 7

Solution: Use $y = mx + b$. Substitute (2,17) and $m = 5$: $17 = 5(2) + b$, so $b = 7$.

Question 20

Skill: Student-Produced Response - Percent Change

The price of a tablet increased from \$240 to \$300. What was the percent increase?

Answer: 25

Solution: Increase = $300 - 240 = 60$. Percent increase = $60/240 \times 100 = 25$.

Question 21

Skill: Student-Produced Response - Geometry

A right triangle has legs of lengths 20 and 21. What is the length of the hypotenuse?

Answer: 29

Solution: By the Pythagorean theorem, $c^2 = 20^2 + 21^2 = 400 + 441 = 841$, so $c = 29$.

Question 22

Skill: Student-Produced Response - Functions

For the function $f(x) = 2x^2 - 3x + 1$, what is $f(4)$?

Answer: 21

Solution: $f(4) = 2(4^2) - 3(4) + 1 = 32 - 12 + 1 = 21$.

Answer Key

SECTION 1, MODULE 1: Reading and Writing

1. B	2. D	3. A	4. C	5. B	6. D	7. A	8. C	9. B
10. D	11. A	12. C	13. B	14. D	15. A	16. C	17. B	18. D
19. A	20. C	21. B	22. D	23. A	24. C	25. B	26. D	27. A

SECTION 1, MODULE 2: Reading and Writing

1. C	2. A	3. D	4. B	5. C	6. A	7. D	8. B	9. C
10. A	11. D	12. B	13. C	14. A	15. D	16. B	17. C	18. A
19. D	20. B	21. C	22. A	23. D	24. B	25. C	26. A	27. D

SECTION 2, MODULE 1: Math

1. B	2. C	3. A	4. D	5. B	6. C	7. A	8. D	9. B
10. C	11. A	12. D	13. B	14. C	15. A	16. D	17. B	18. 6
19. 70	20. 11	21. 72	22. 5					

SECTION 2, MODULE 2: Math

1. C	2. B	3. D	4. A	5. C	6. B	7. D	8. A	9. C
10. B	11. D	12. A	13. C	14. B	15. D	16. A	17. C	18. 6
19. 7	20. 25	21. 29	22. 21					